

Energy Optimization, Characterisation and Process Simulation of Crude Oil Distillation in an Atmospheric Distillation Column

A Process Simulation Report

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Abstract

This report presents the characterisation of a crude oil feedstock and the steady-state process simulation of its separation in an atmospheric distillation column. The crude oil is characterised using assay data comprising light-ends composition, a true boiling point (TBP) distillation curve, and bulk physical properties. The conditioning train, consisting of pre-heating, desalting, and heat-recovery stages, is described together with the design and operating basis of the main atmospheric column, which produces naphtha, kerosene, diesel, and automotive gas oil (AGO) as principal products alongside an atmospheric residue. Column internals, side-draw and pump-around configurations, and key operating specifications are summarised. The results provide a consistent basis for evaluating column performance and product yields under the specified operating conditions.

Contents

1	Introduction	3
2	Crude Oil Feed Characterisation	3
2.1	Bulk Properties and Feed Conditions	3
2.2	Light Ends Composition	3
2.3	True Boiling Point (TBP) Distillation Data	4
3	Process Description	4
3.1	Overview	4
3.2	Crude Conditioning Train	5
4	Atmospheric Distillation Column	5
4.1	Column Configuration	5
4.2	Product Side Streams	6
4.3	Pump-Around Circuits	6
4.4	Overhead and Distillate Specifications	7
5	Simulation Figures	7
6	Results and Discussion	11
7	Conclusion	11

1 Introduction

Crude oil is a complex mixture of hydrocarbons that must be separated into fractions of commercial value before further downstream processing. The primary separation step in a refinery is carried out in an atmospheric distillation column, where the crude is fractionated on the basis of relative volatility into light gases, naphtha, kerosene, diesel, automotive gas oil (AGO), and atmospheric residue.

Accurate simulation of this separation requires proper characterisation of the crude feedstock. Since crude oil cannot be represented as a set of discrete pure components, it is described using an assay consisting of a light-ends composition, a true boiling point (TBP) distillation curve, and bulk physical properties such as density. This information is used by the process simulator to internally generate pseudo-components with corresponding boiling point, molecular weight, density, and viscosity distributions that represent the crude.

This report documents the feed characterisation, the process flow configuration of the crude preheat and conditioning train, and the design and operating specifications of the atmospheric distillation column used to separate the crude into its principal products.

2 Crude Oil Feed Characterisation

2.1 Bulk Properties and Feed Conditions

The crude oil feed is characterised by the bulk standard density and the inlet stream conditions summarised in Table 1.

Table 1: Crude oil feed stream conditions

Parameter	Value
Temperature	15 °C
Pressure	1000 kPa
Mass flow rate	6×10^5 kg/h
Bulk standard density	879.8 kg/m ³

A desalting water stream is also fed to the process at 15 °C with a mass flow rate of 21 600 kg/h, used to wash soluble salts and solids from the crude prior to preheating.

2.2 Light Ends Composition

Light ends are defined as the pure, low-boiling-point components present in the crude, ranging from C2 to C5. Their liquid volume percentages, as obtained from the crude assay, are given in Table 2.

Table 2: Light ends composition of the crude assay (liquid volume %)

Component	Composition (vol.%)
Methane	0.0065
Ethane	0.0225
Propane	0.3200
i-Butane	0.2400
n-Butane	1.7500
i-Pentane	1.6500
n-Pentane	2.2500

2.3 True Boiling Point (TBP) Distillation Data

The true boiling point distillation curve, expressed as cumulative liquid volume percent against boiling temperature, is used to generate the internal pseudo-component distribution for the crude. The assay data are presented in Table 3.

Table 3: True boiling point (TBP) distillation assay data

Cumulative Liquid Volume (%)	Temperature (°C)
0.00	−12
4.00	32
9.00	74
14.00	116
20.00	154
30.00	224
40.00	273
50.00	327
60.00	393
70.00	450
76.00	490
80.00	516

The simulator uses this assay information (light ends, TBP curve, and bulk density) to internally fit continuous TBP, molecular weight, density, and viscosity curves, from which a set of pseudo-components is generated to represent the crude oil for phase-equilibrium and property calculations.

3 Process Description

3.1 Overview

The overall process consists of a crude conditioning train followed by fractionation in an atmospheric distillation column. The conditioning train raises the crude temperature in stages, using heat recovery from internal pump-around streams, before final heating in a furnace and delivery to the column. Figure 1 shows the overall process flow diagram.

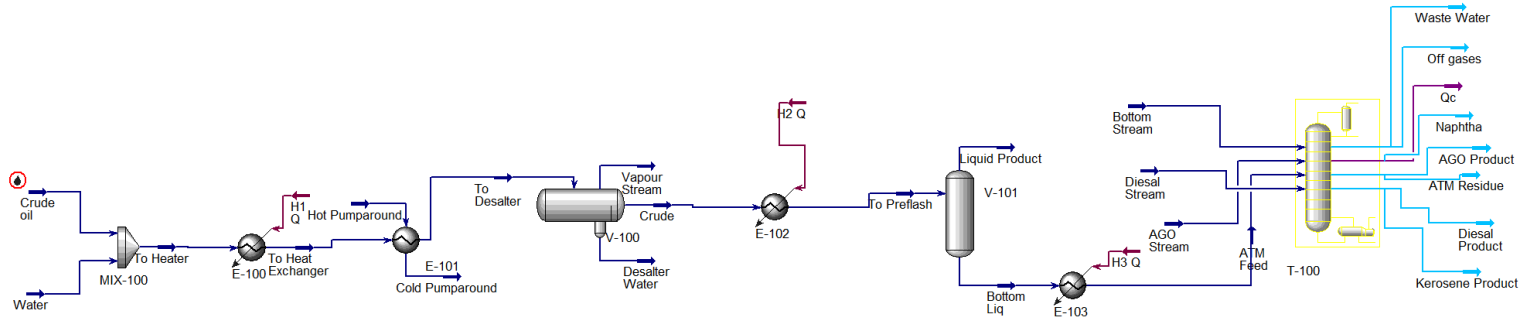


Figure 1: Process flow diagram of the crude oil conditioning train and atmospheric distillation column.

3.2 Crude Conditioning Train

The raw crude and desalting water are combined in a mixer (MIX-100) prior to initial heating. The main conditioning steps are summarised below.

- **Simple Heater 1:** The mixed crude/water stream is heated with a pressure drop of 50 kPa, raising the stream temperature to 65 °C.
- **Shell-and-tube Heat Exchanger:** The crude is preheated against the hot pump-around return stream. The tube side (crude) is heated to 180 °C at 200 kPa, with a standard ideal liquid volume flow of 175 m³/h and a minimum approach temperature of 30 °C between the hot and cold streams.
- **Desalter:** The preheated crude is sent to a desalter vessel, where the wash water and entrained salts/solids are removed as a desalter water stream.
- **Simple Heater 2:** The desalted crude is further heated, with a pressure drop of 375 kPa, to a pre-flash temperature of 175 °C.
- **Pre-Flash Drum:** Light components are separated from the crude prior to the final furnace heating stage.
- **Furnace:** The bottom liquid from the pre-flash drum is heated in a furnace, with a pressure drop of 250 kPa, to a final feed temperature of 400 °C before entering the atmospheric column.

4 Atmospheric Distillation Column

4.1 Column Configuration

The atmospheric column is modelled as a reflux absorber with 29 stages. The heated crude feed enters at stage 28, and the furnace bottom liquid enters as the atmospheric residue feed. Table 4 summarises the principal column design and operating parameters.

Table 4: Atmospheric distillation column design and operating parameters

Parameter	Value
Number of stages	29
Feed stage	28
Bottom liquid outlet	Atmospheric residue
Bottom stream vapour fraction	1
Bottom stream pressure	1380 kPa
Bottom stream flow rate	3400 kg/h
Condenser pressure	140.0 kPa
Condenser pressure drop	60.0 kPa
Bottom stage pressure	230.0 kPa
Condenser temperature	40 °C
Top stage temperature	120 °C
Bottom stage temperature	340 °C

4.2 Product Side Streams

Three liquid side products – kerosene, diesel, and automotive gas oil (AGO) – are withdrawn from the column at intermediate stages, with kerosene reboiled and the diesel and AGO streams steam-stripped. Table 5 lists the draw and return stages, stripping configuration, and product flow rate for each side stream, and Table 6 lists the additional temperature, pressure, and flow specifications applied to the diesel and AGO product streams.

Table 5: Side-stream draw configuration

Parameter	AGO	Diesel	Kerosene
Draw stage	22	17	9
Return stage	21	16	8
Configuration	Steam stripped	Steam stripped	Reboiled
Product flow rate	30.69 m ³ /h	128.88 m ³ /h	64.10 m ³ /h

Table 6: Diesel and AGO product stream specifications

Parameter	Diesel Stream	AGO Stream
Temperature	150 °C	150 °C
Pressure	350 kPa	350 kPa
Flow rate	1350 kg/h	1150 kg/h

4.3 Pump-Around Circuits

Pump-around circuits are used to recover heat from the column at three locations, improving thermal efficiency and controlling internal vapour/liquid traffic. Table 7 summarises the draw/return stages, circulation flow rates, and heat duties for each pump-around.

Table 7: Pump-around circuit specifications

Parameter	AGO PA	Diesel PA	Kerosene PA
Draw stage	22	17	9
Return stage	21	16	8
Flow rate	200 m ³ /h	200 m ³ /h	330 m ³ /h
Duty	-3.7×10^7 kJ/h	-3.7×10^7 kJ/h	-4.5×10^7 kJ/h

4.4 Overhead and Distillate Specifications

The column overhead is fully condensed, with the following specifications applied at the top of the column:

Table 8: Overhead product specifications

Parameter	Value
Overhead vapour rate	0 kgmol/h
Distillate (naphtha) rate	104.3 m ³ /h

5 Simulation Figures

This section presents the profiles generated from the converged column simulation, showing how temperature, pressure, vapour/liquid traffic, composition, and physical properties vary from the top tray (stage 1) to the bottom of the column (around stage 29), together with the light-key K-values used in the fractionation calculation. The stripper sections attached to the kerosene, diesel, and AGO side draws appear as the additional stages beyond stage 29 in these profiles.

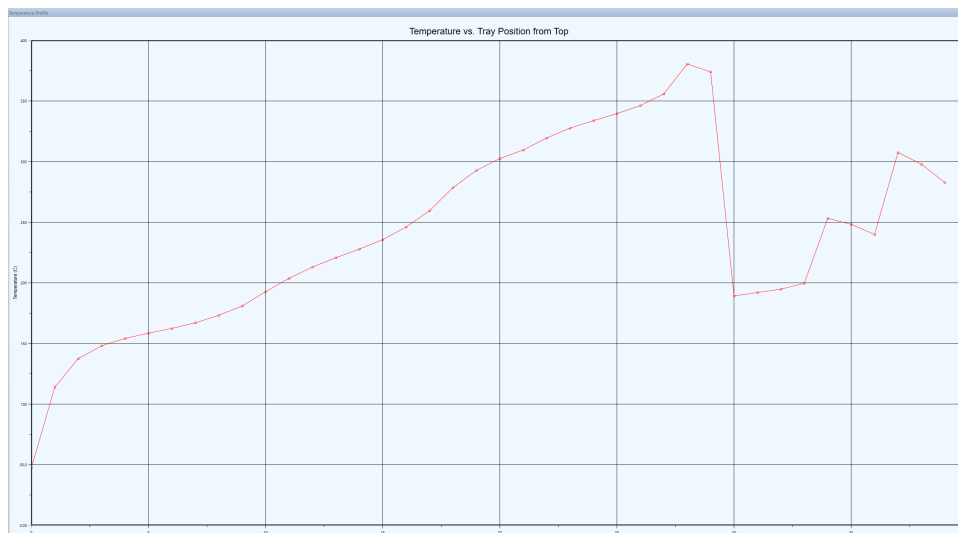


Figure 2: Column temperature profile as a function of stage number.

The temperature profile rises steadily from about 50 °C at the top tray to a maximum of approximately 380 °C in the flash zone around stage 27–28, consistent with the 400 °C furnace feed temperature entering the column at stage 28. The temperature then drops sharply at the bottom stages, reflecting the transition into the stripping/side-draw sections, before rising again

in the individual side-stripper columns (stages beyond 29) as the kerosene, diesel, and AGO cuts are steam-stripped.

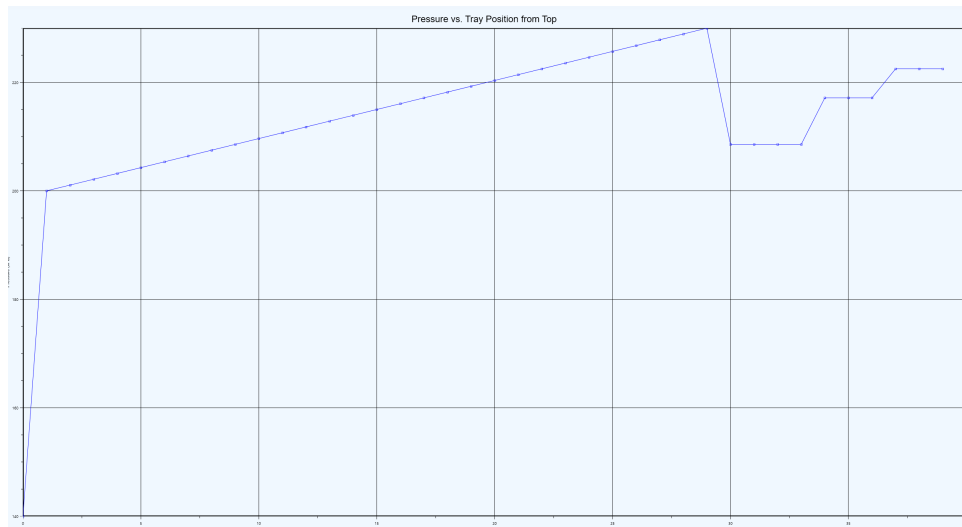


Figure 3: Column pressure profile as a function of stage number.

The pressure profile increases nearly linearly from the condenser pressure of 140 kPa at the top stage to approximately 230 kPa at the bottom stage, consistent with the specified condenser and bottom-stage pressures in Table 4. The pressure drop across the side strippers is small in comparison, as reflected by the near-constant pressure levels in the corresponding stages beyond stage 29.

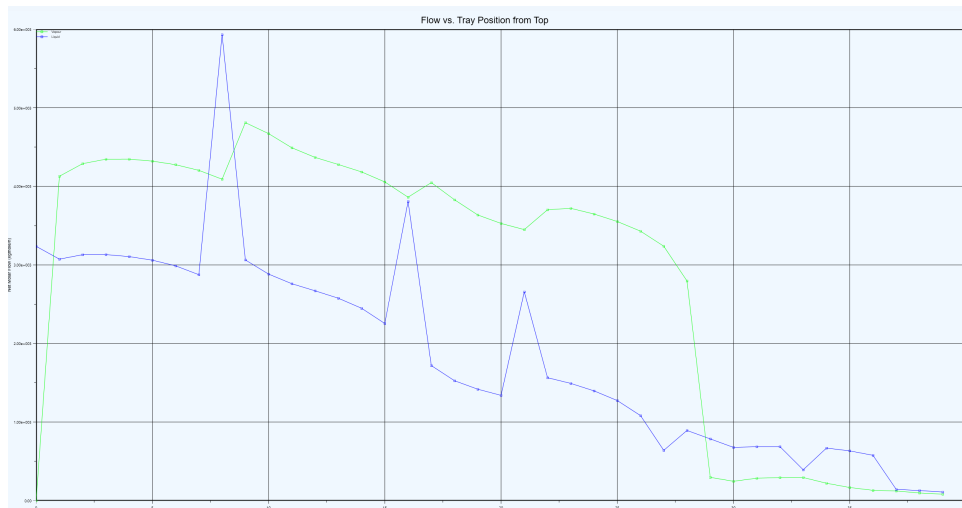


Figure 4: Net vapour and liquid molar flow profiles as a function of stage number.

The vapour and liquid traffic profiles in Figure 4 show the internal molar flows within the column. The liquid flow decreases progressively down the main column as material is withdrawn at each side-draw stage, with sharp local spikes corresponding to the return of stripped vapour and liquid from the kerosene, diesel, and AGO strippers back into the main column. The vapour flow rises quickly near the top of the column, associated with the reflux and light-ends flash, then gradually declines through the lower stages before falling sharply into the stripping sections, where vapour rates are comparatively small.



Figure 5: Light-ends liquid-phase composition profile (methane, ethane, propane, and i-butane) as a function of stage number.

The light-ends composition profile confirms that methane, ethane, propane, and i-butane are concentrated almost entirely in the liquid phase at the top few stages of the column, with mole fractions falling essentially to zero beyond approximately stage 5. This is consistent with these components being flashed overhead and recovered predominantly in the vapour distillate rather than remaining in the descending liquid.

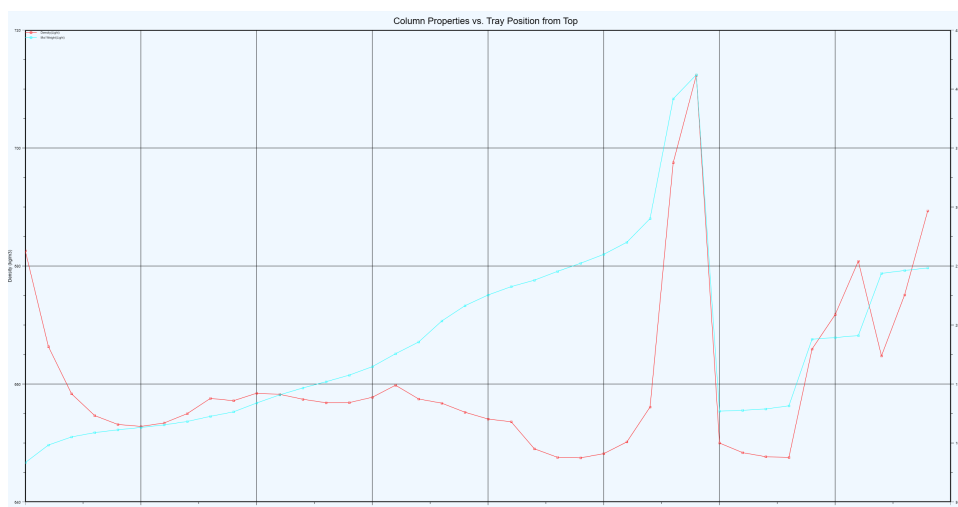


Figure 6: Liquid density and molecular weight profile as a function of stage number.

Figure 6 shows the liquid molecular weight increasing steadily from the top of the column to the flash zone, reflecting the progressively heavier hydrocarbon fractions accumulating toward the bottom trays, with a sharp peak in both density and molecular weight around stage 28–29 corresponding to the heavy atmospheric residue material near the feed and bottom stages. The corresponding density profile follows a broadly similar trend, with local variations at the side-draw stages reflecting the differing character of the kerosene, diesel, and AGO cuts.

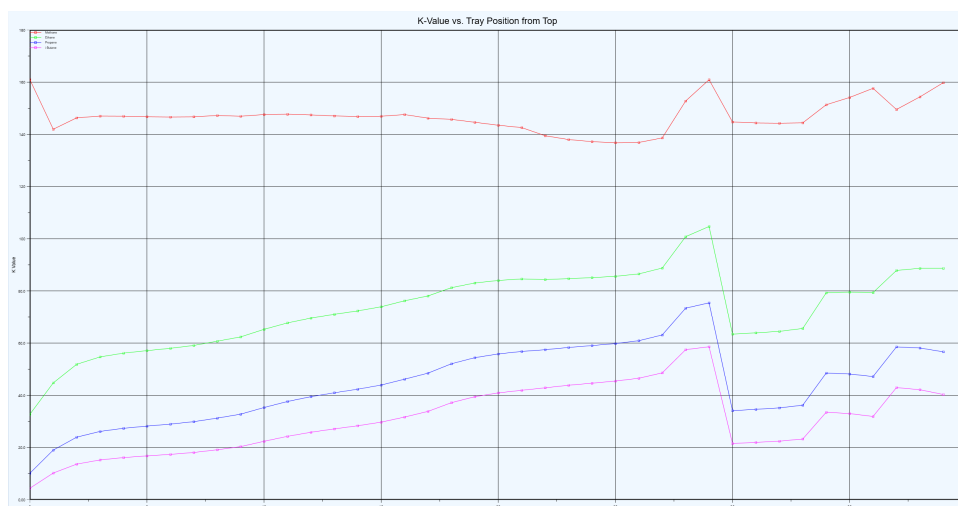


Figure 7: K-values of methane, ethane, propane, and i-butane as a function of stage number.

The K-value profiles in Figure 7 show methane maintaining the highest volatility throughout the column, as expected, while ethane, propane, and i-butane K-values increase steadily down the main column as temperature rises, before dropping abruptly at the transition into the side strippers, where the lower operating temperatures of the stripping sections reduce the relative volatility of these light components.

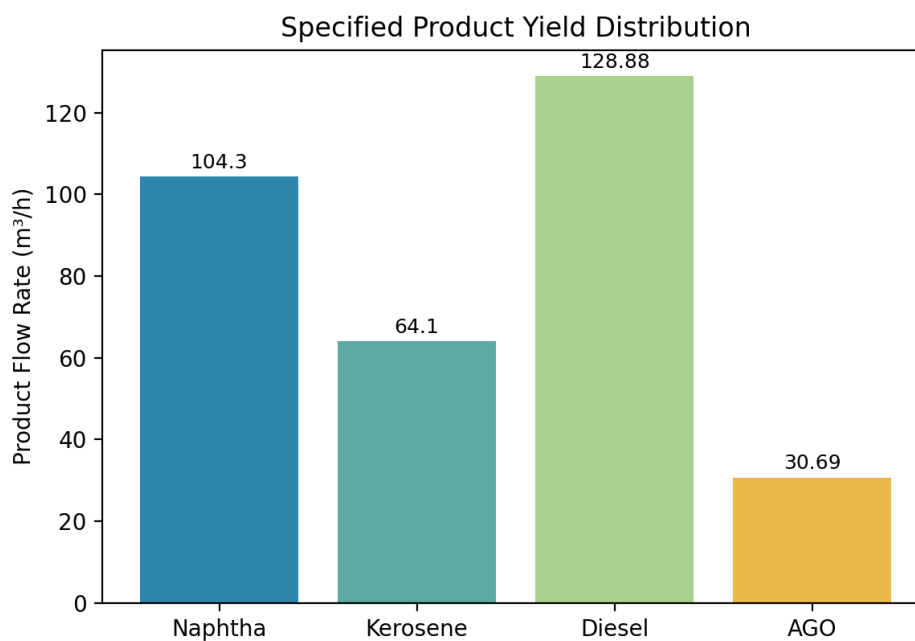


Figure 8: Product yield distribution by mass or volume flow.

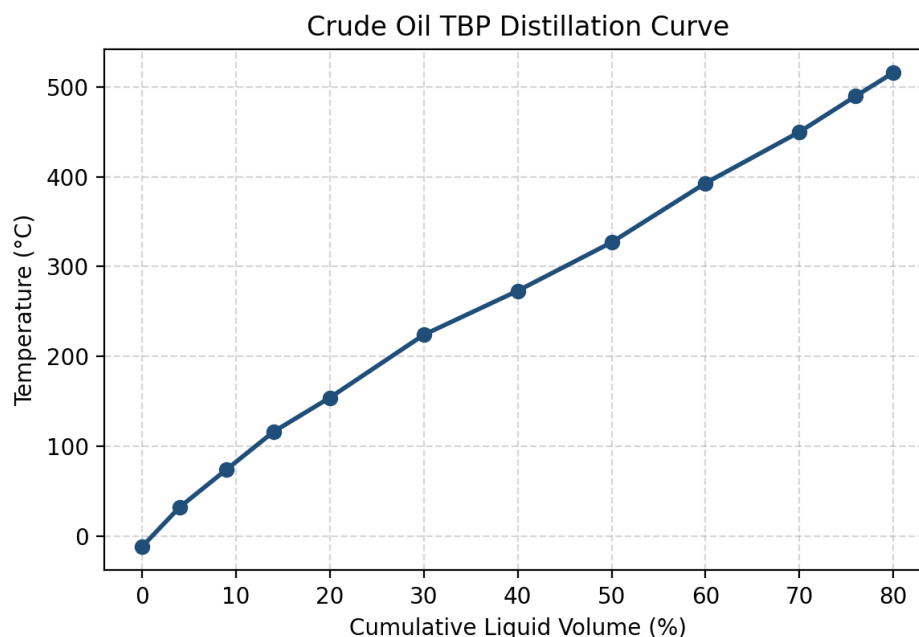


Figure 9: True boiling point (TBP) distillation curve of the crude feed.

6 Results and Discussion

The process is designed to separate the crude oil into five principal streams: light ends/off-gas, naphtha, kerosene, diesel, AGO, and atmospheric residue. The preheat train raises the crude temperature from 15 °C to 400 °C across four heating stages (Simple Heater 1, the crude/pump-around heat exchanger, Simple Heater 2, and the furnace), recovering heat internally from the hot pump-around streams before final firing in the furnace.

Within the atmospheric column, the three pump-around circuits remove a combined duty of approximately 9.9×10^7 kJ/h, which both controls the internal vapour load and improves overall energy efficiency by reducing the condenser duty. The side-draw configuration – with kerosene reboiled and diesel/AGO steam-stripped – reflects standard industrial practice for adjusting the flash point and stripping light components from each product cut.

The simulated column profiles presented in Section 5 above are internally consistent with this operating basis: the temperature profile tracks the specified furnace feed and condenser temperatures, the pressure profile matches the specified top- and bottom-stage pressures, and the light-ends composition and K-value profiles confirm that the light components are effectively rejected to the column overhead. The combined liquid product flow rate from the three side streams and the atmospheric residue, together with the specified naphtha distillate rate of 104.3 m³/h, define the overall product yield distribution of the column under the specified operating conditions. These results form a consistent basis for further sensitivity analysis, such as evaluating the impact of pump-around duty, feed temperature, or column pressure on product yields and quality.

7 Conclusion

This report presented the characterisation of a crude oil feedstock and the process configuration used to simulate its separation in an atmospheric distillation column. The crude was char-

acterised using light-ends composition, a TBP distillation curve, and bulk density data. The conditioning train – comprising staged heating, heat recovery, and desalting – prepares the crude for fractionation, while the atmospheric column, operating with three pump-around circuits and steam-stripped/reboiled side draws, separates the crude into naphtha, kerosene, diesel, AGO, and atmospheric residue. The simulated temperature, pressure, flow, composition, density, and K-value profiles across the column stages are consistent with the specified design and operating basis, and together provide a complete and self-consistent basis for reproducing the simulation and for further process optimisation studies.